

The impacts of climate policy uncertainty on return, volatility, correlation and tail dependence of China's and US stock markets

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Abstract: This paper measures China's daily and monthly climate policy uncertainty (CPU) from Jan 2000 to Mar 2022 based on Chinese news data for the first time. Then, the nonlinear and lag impacts of the US CPU and China's CPU on the return, volatility, correlation and tail dependence of China's and US stock markets are investigated and compared by adopting GARCH (1,1), copula function and the distribution lag nonlinear model (DLNM). The data of stock markets includes the Shanghai Composite Index (SSCI) and NASDAQ from Jan 2000 to Mar 2022 from the Choice database, and the Shenzhen Composite Index (SCI) and S&P 500 are used for the robustness test. The empirical results indicate that (1) the growth trend of China's CPU index is similar to that of the US. However, there are significant differences between the impacts of these two CPUs on stock markets. (2) For China, high CPU decreases current stock market return and increases volatility but decreases it in the future. It could also increase the upper tail dependence between China's and the US stock markets' volatilities in current period. (3) For the US, CPU decreases stock market return in the short term but increases it in the long term. High CPU increases volatility in short term, decreases volatility in 5 months and increases it again after 6 months. Both low and high CPU could increase the correlation between China's and US stock markets' volatilities.

Keywords: Climate policy uncertainty; Stock market volatility; Distribution lag nonlinear model; Upper tail dependence; Nonlinear and lag effects

JEL Codes: Q54, C43, F30, F36

1 Introduction

Climate change is one of the most serious challenges faced by human beings in the 21st century and also is the focus of current governments (Koh et al., 2021; Boulange et al., 2021). According to 'Climate Change 2022: Impacts, Adaptation and Vulnerability', issued by the Intergovernmental Panel on Climate Change (IPCC), the impacts and risks of climate change are increasingly complex

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and difficult to manage. Multiple climate risks and nonclimate risks will interact, leading to multiple climate disasters occurring simultaneously, such as droughts, typhoons, fires and floods (Boulange et al., 2021). It also could lead to varying degrees of negative impacts on economic growth, employment, agricultural losses, renewable energy consumption and population health (Dafermos et al., 2018; Galaz et al., 2018; Yang, 2019; Solaun and Cerdá, 2019; Oremus, 2019).

In addition to the above effects, based on climate economics and financial vulnerability theory, the impact of climate change on the economy and residents' lives will be transmitted to the financial system through the real economy, so governments and financial institutions are increasingly aware of the need to take climate and environmental risks into consideration in the risk assessment process of the financial system (Battiston et al., 2017). Due to the advancement of globalization, countries have shown strong linkage effects in trade, investment, production and finance, and the financial systems of countries have significant contagion characteristics. The financial risks caused by climate change will further expand (Bardoscia et al., 2018). Governments, central banks and financial regulators are beginning to recognize that they have an active role to play in tackling climate change. In December 2015, the Financial Stability Board (FSB) of the G20 set up the Task Force on Climate-related Financial Disclosure (TCFD). The group divides climate change risks into physical and transformational risks. Physical risks include all kinds of natural disasters and events related to the environment and climate, while transition risks include policy and legal risks, technological risks, market risks and goodwill risks.

To protect against the risks of climate change, various policies to combat climate change have been published, and they have the potential to cause economic and financial shocks to varying degrees (Dafermos et al., 2018; Paroussos et al., 2019). For example, the low-carbon transition policy may cause the share price of fossil energy companies to fall, while the subsidy policy of new energy vehicles may cause the share price of related energy-saving car companies to rise. Uncertainty over transformational policies to address climate change could raise equity risk, as entrepreneurs who do not know whether future policies will be aggressive or conservative can do little more than wait and see and may scale back their long-term investments. Some expansionist companies' blind investment in climate policies also will increase financial risks. In addition, policies to deal with

sudden large-scale climate disasters can influence financial markets and cause financial risks by means of transfer payments or compulsory administrative orders (Stevanović et al., 2016; Landis and Bernauer, 2012).

Although climate policy uncertainty is an important factor in the volatility of China's stock market, due to the lack of accurate measurement of climate policy uncertainty, it is difficult to effectively capture its impact, including its nonlinear and lag effects on stock market volatility. Furthermore, for different countries, such as China and the US, climate policy uncertainty may have different impacts, and this comparison has not been involved in existing research. Considering the volatility spillover effect in international stock markets, the influence of the correlation between different stock markets also should be valued.

Therefore, compared with previous studies, there are two main innovations in this paper. First, based on the methods used in Huang and Luk (2019), which measure economic policy uncertainty, this paper measures the level of climate policy uncertainty in China for the first time. In contrast to the vocabulary adopted in Huang and Luk (2019), the vocabulary about policy and uncertainty are added, and this paper proposes a new collection of words about climate change, environmental governance and low-carbon transition on the basis of Li et al. (2020), which could provide a reference for research on climate policy uncertainty. Specifically, we provide a daily and monthly index of China's climate policy uncertainty from January 1, 2000, to March 7, 2022. The time trend chart of the index is similar to that of the US reported in Li et al. (2020). Climate policy uncertainty is similar to economic policy uncertainty, which can better reflect the fluctuations of climate policy and help predict the development prospects of related industries and technologies, such as new energy vehicles, the photovoltaic industry, clean energy, technical emission reduction technology and green bonds. In this paper, higher the daily and monthly CPU values indicates the higher the uncertainty.

Second, this paper confirms that climate policy uncertainty has significant, nonlinear and lag impacts on stock market volatility and that these impacts are different under different levels of climate policy uncertainty. The intensity of influence corresponding to different lag periods also is

reported. The differences in these complex impacts of climate policy uncertainty on stock market volatility in China and the US are compared for further analysis. The complex impacts of CPU on stock market return, volatility, correlation and upper tail dependence in China and the US are analyzed and compared. We confirmed that for China, high CPU decreases current stock market return and increases volatility but decreases it in the future. However, for the US, CPU decreases stock market return in the short term but increases it in the long term. Beyond that, empirical results also shown that both low and high CPU in the US could increase the correlation between China's and US stock markets' volatilities.

The rest of this paper is structured as follows. Section 2 presents a review of the literature. Section 3 explains the data and methods. Section 4 reports the empirical results and robustness test. Section 5 presents the conclusions.

2 Literature review

This part first sorts out the research on the correlation of stock market risk and the impact of climate change on economy and finance, and then we review relevant studies on policy uncertainty assessment. Finally, the current research on climate policy from the perspective of policy uncertainty is sorted out and analyzed.

2.1 Research on the correlation of stock market risk

The correlation of stock market risk is mainly due to risk correlation caused by economic correlation and risk correlation caused by investor behavior (Litimi et al., 2016; Chang et al., 2020). Specifically, they include international trade linkages (Corea and Radev, 2014), capital market and money market linkages (Pagano and Sedunov, 2016), transnational credit (Bastos and Pindado, 2013), international investment (Cipriani et al., 2013), industrial linkages (Tashjian et al., 2016), investor expectation effects (Ahmad and Oriani, 2022), and leverage and debt correlations (Drehmann and Juselius, 2014).

In terms of empirical research, most of the studies use econometric models, complex network methods, signal decomposition technology, input–output analysis and information entropy to analyze the correlation between stock market risk and stock market volatility (Dimpfl and Peter, 2014; Yang et al., 2016; Wang and Hui, 2017). Liu et al. (2017) used the BEKK–GARCH model and stock market data of G20 countries to construct a regional stock market network and analyze its volatility spillover relationship. Silva et al. (2016) studied the evolution of global financial market network topology and found that the network was relatively fragile before the 2008 financial crisis but subsequently became more resilient. It provides important theoretical support for revealing the law of financial risk dynamic contagion and preventing financial risk contagion. Bardoscia et al. (2018) found that the topological characteristics of network structure would affect the ease of crisis spreading in the system and analyzed the stability process of the financial system, pointing out that market integration and diversification would form a cyclical structure, which would further promote the instability of the financial system, thus exacerbating the financial crisis.

2.2 Economic and financial impacts of climate change

Many studies have confirmed that climate change has a significant impact on the economy, social production, energy and people's lives (Koh et al., 2021; Solaun and Cerdá, 2019). Specifically, regarding the impact of climate change on the economy, Pretis (2020) believed that the estimation of the impact of climate change on the economy is crucial for policy decisions and proved that the climate energy balance model is equivalent to an econometric cointegration system, which can be estimated in discrete time. The estimated parameters can be used to quantify the uncertainties in a comprehensive assessment model of the economic impacts of climate change. Calel et al. (2020) pointed out that many assessments of the economic impacts of climate change rely only on a small number of climate–economic coupling models. These assessments account for the economic costs of cognitive uncertainty stemming from the inability to accurately estimate key model parameters, such as equilibrium climate sensitivity. Therefore, we must pay attention to the analysis of uncertainty in dealing with climate change. In view of the impact of climate change on energy, Solaun and Cerda (2019) sorted out relevant studies on the response of solar, wind, hydraulic and other renewable energy power generation technologies to the impact of climate change from the

perspective of quantitative analysis and pointed out that research on the impact of climate change on renewable energy is becoming increasingly important. Yang (2019) used numerical simulation to study the relationship between energy-intensive industries and climate change, providing a reference for regional climate negotiations from the perspective of externalities. In view of the impact of climate change on fisheries, Oremus et al. (2019) adopted the North Atlantic Oscillation Index (NAO) as a proxy variable of climate change to analyze the impact of climate change on fisheries in England. It was found that climate shocks not only reduced regional catches and incomes in the new sector but also reduced county wages and employment in related industries. For each standard deviation increase in climate mean, county-level fishery employment decreased by 13% on average. Holsman et al. (2020) evaluated the management strategies of major US fisheries in the eastern Bering Sea. The study found that climate change has significant negative impacts on fisheries, and climate-driven changes may exceed the adaptive capacity of current fisheries management after 2050. In view of the impact of climate change on social welfare, Yan et al. (2016) constructed the Bergson–Samuelson social welfare function based on the basic concepts of climate change science and welfare economics theory to evaluate social and economic vulnerability and economic welfare risk under the background of climate change. In addition, the economic losses and welfare risks of climate disasters under climate change scenarios in China from 2016 to 2030 are estimated by a regional weighted method, a special national adaptation fund must be established that is designed in an overall and forward-looking way from the strategic perspective of climate equity and climate security, and adaptation resources and inputs must be prioritized for the most vulnerable regions to meet their basic needs and promote long-term sustainability. Stevanović et al. (2016) pointed out that climate change threatens global agricultural productivity and leads to rising food prices.

In addition to the above impacts, some scholars also have noted the financial risks caused by climate change. As one of the major factors leading to structural changes in the financial system, climate change is long-term, structural and overall is attracting the attention of global financial institutions and investors (Dietz et al., 2016). Financial risks caused by climate change can be generally divided into two categories: (1) physical risks, namely, financial risks caused by failure to effectively solve the problem of climate change, such as the losses of residents and enterprises caused by large-scale

climate disasters, which ultimately affect the stock market (Hirabayashi et al., 2013); and (2) transformation risks, namely, the risk of financial system inadaptability caused by effective policies and actions taken by public or private sectors to control climate change (Guilhot, 2022). These two types of risks will have a significant impact on macroeconomic and financial variables through channels such as asset revaluation, balance sheets, collateral value changes, exposure to risk positions, policy uncertainty and expected market volatility, and then impact financial stability and the macroeconomy.

Under the financial accelerator and collateral constraint mechanisms, market signals may amplify the severity of climate-related risks, making their impact on individual financial institutions evolve into systemic risks (Klomp, 2014). Beyond that, the impact of climate risk on the financial system and the macroeconomy is characterized by 'cyclic feedback', which has a wider and more far-reaching impact than other risks (Coeure, 2018). Losses from climate disasters will lead to a contraction in credit, which will further weaken household and corporate balance sheets, affecting potential growth and the output gap (Bos et al., 2022). The intrinsic externality of climate change leads to the failure of market price signals, which requires proper intervention and guidance by the "visible hand" of the government (Dafermos et al., 2018). In view of the possible systemic impact of climate change on the financial system, central banks, as macropolicy managers, should be forward-looking enough to effectively deal with climate-related risks (Battiston et al., 2017). Monasterolo et al. (2019) pointed out that traditional climate economics and financial risk models cannot well consider the characteristics of climate risk and the opportunities brought about by climate change due to the constraints of equilibrium conditions and linear effects as well as representative factors and intertemporal optimization. At present, research in this field urgently needs a new model that incorporates the uncertainties and complexities of climate impacts on socioeconomic systems and can consider the uncertainties and complexities of climate responses to socioeconomic systems. In this regard, approaches rooted in evolutionary economics and complexity science can provide complementary insights to traditional climate economic models. Campiglio et al. (2018) believed that climate change poses important challenges for central banks and financial institutions, and that current assessments of climate-related financial risks are hampered by various challenges. First, the data needed to conduct a comprehensive climate stress

test are often nonexistent or too low-resolution for researchers outside financial regulators to access. Second, a comprehensive assessment of climate-related financial risks cannot rely on static snapshots alone: it needs to model the dynamic interactions among the macroeconomy, the financial system, climate change and environmental policies.

2.3 Climate policy and the measurement of its uncertainty

To avoid climate change risks, respond to climate disasters, promote green and sustainable economic growth and ensure the safety of people's work and life, the government will issue a series of climate policies in a timely manner (Chen et al., 2021; Wesseh et al., 2022). These policies are inherently uncertain and also will affect the uncertainty of the economy and stock market (Coeure, 2018). As the stock market is characterized by obvious risk contagion, the financial risks caused by climate change will continue to spread (Dietz et al., 2016).

In fact, the impact of climate policy uncertainty on the stock market is an important realization path of climate change transition risk (Kunreuther et al., 2013). Relevant government departments will issue relevant policies to address climate risks from time to time based on the impacts of climate change mentioned above (Kunreuther et al., 2013; Stern et al., 2013; Paroussos et al., 2019). These measures include, but are not limited to, low-carbon transition policies (Guilhot, 2022; Wang and Yang, 2022; Dugan et al., 2022), renewable energy subsidies (Liu and Ronn, 2020; Bai et al., 2021; Qi et al., 2022; Zhang et al., 2022), carbon taxes (Cai et al., 2015; Cui et al., 2021), green finance (Madaleno, 2022), encouraging green consumption and travel methods (Lin et al., 2021), climate-related transfer subsidies (Landis and Bernauer, 2012), development of low-carbon energy technologies (Helveston and Nahm, 2019), water conservancy facilities planning (Fletcher et al., 2019), etc. The release of these climate policies will have complex economic and financial implications.

Batten (2018) argues that unexpected policy transitions may trigger negative macroeconomic shocks from the supply side. A sudden policy shift, for example, in the form of higher carbon taxes on fossil fuels and outright quantitative limits, would sharply raise energy prices and thus undermine global

economic growth. McGlade et al. (2015) estimated that 35% of the existing global oil reserves, 52% of natural gas reserves and 88% of coal reserves would become unusable by 2050 to keep global warming below 2 °C without the use of carbon dioxide capture and storage technology. Taking these assets off the balance sheets of fossil fuel companies has a very negative impact on corporate value. In addition, in the absence of market expectations, many assets will be revalued in the short term, which will easily lead to cyclical losses and long-term tightening of financial conditions (Batten, 2018). Coeure (2018) found that a sudden tightening of the carbon emission policy could lead to a disorderly repricing of carbon-intensive assets, which would shock the price of fossil fuels and the value of the relevant enterprises, in turn affecting their solvency and the financial position of investors holding their debt or equity, and even cause systemic damage to the entire financial system.

Most existing studies focus on the economic and financial impacts of a climate policy, but the analysis from the perspective of policy uncertainty deserves more attention, and that was involved in only a few studies. The relevant studies focus more on climate decision-making in the context of uncertainty (Workman et al., 2021) or uncertainty of the two types of climate change risks (Chenet et al., 2021). In a recent study, Gavrilidis (2021) measured the uncertainty of US climate policy according to the news data of major US newspapers and found that the uncertainty of climate policy has a strong negative impact on carbon dioxide emissions. Then, based on this data, Bouri et al. (2022) analyzed its impact on green stocks. These studies provide good enlightenment. To our knowledge, no studies have measured the daily climate policy uncertainty index in China. If climate policy uncertainty cannot be accurately measured, it is difficult to quantitatively analyze its impacts on the economy or stock market with the help of regression models.

In summary, the sudden release of a climate policy or an emergency executive order in response to a major climate disaster tends to have a potentially delayed impact on the economy, which in turn affects stock market volatility. Uncertainty over various climate-related policies issued to address climate change can easily lead to economic and financial market turmoil in this way. In addition, due to the characteristics of significant risk correlation between stock markets, the climate policy issued by one country is likely to affect the correlation with stock market fluctuations of neighboring countries or regions. In addition, the impact of such policies on the macroeconomy and the stock

market has a lag and nonlinear effect, which may be different under different policies or different national scenarios. Therefore, it is necessary to quantitatively analyze the impact of climate policy uncertainty on stock market volatility and volatility correlation from the perspective of hysteresis and nonlinearity.

3 Data, measurement of China's CPU and regression methods

3.1 Data sources

3.1.1 Data for calculating the China's CPU

At present, index measurement of financial market has been involved in many studies, but uncertainty measurement of climate and environment is still a difficulty. Most relevant studies in this area are based on news media data (Lucey, et al., 2022; Wang et al., 2022). This paper collects all news involving climate policy uncertainty from 9 main Chinese newspapers from January 1 2000 to February 17, 2022. We refer to Huang and Luk (2018) in the selection of newspapers, and the newspapers include *Beijing Youth Daily*, *Guangzhou Daily*, *Jiefang Daily*, *People's Daily Overseas Edition*, *Shanghai Morning Post*, *Southern Metropolis Daily*, *The Beijing News*, *Wen Hui Daily* and *Yangcheng Evening News*. *Today Evening Post* is excluded because the number of related articles was very low during 2000-2019. Based on Huang and Luk (2018), the sum of the number of related articles appearing each month is calculated as a monthly index. Beyond that, we also report a daily index of climate policy uncertainty, even though these articles are very scarce for each day. Similar to Huang and Luk (2018), all data come from the Wisers Information Portal. These newspapers focus on important policies in major Chinese cities.

3.1.2 Data for the stock market

This paper adopts the volatility of the stock market index in China and the US as the dependent variables. In this section, we collect the log rate of returns of the Shanghai (securities) Composite Index (SSEC), NASDAQ Composite Index, Shenzhen Composite Index (SCI) and S&P 500 from

the Choice database. The closing price of these indices starts in Dec 1999 and ends in Mar 2022; then, the log rate of returns is calculated from Jan 2000 to Mar 2022.

3.2 Measurement of China’s CPU

The keywords for uncertainty refer to Huang and Luk (2018). The keywords for climate change are based on Li et al. (2020), and some additional terms related to low carbon and the green economy have been added. The keywords for policy also are based on Huang and Luk (2018), and some additional terms are added according to the white paper “China’s Policies and Actions on Climate Change” issued by the State Council Information Office of the People’s Republic of China. The relevant Chinese keywords used to search related news are provided in Table 1.

Table 1 Chinese keywords about China’s CPU Index

Criteria	Chinese and English translation
Climate	雹子(hail), 暴风雪(snowstorm), 暴雪(blizzard), 暴雨(rainstorm), 冰雹(hail), 冰暴(ice storm), 采暖季(heating season), 大风(windy), 大气(atmosphere), 大雾(fog), 大雪(heavy snow), 大雨(heavy rain), 低气压(low pressure), 低碳(low carbon), 地震(earthquake), 多云(partly cloudy), 二氧化碳(carbon dioxide), 风暴(storm), 风暴季节(storm season), 风暴损失(storm damage), 风暴影响(storm impact), 风暴云(storm cloud), 风沙(sandstorm), 风雪(snowstorm), 干旱(drought), 干涸(dry up), 哥本哈根(Copenhagen), 供暖季(heating season), 海啸(tsunami), 海震(seaquake), 寒潮(cold wave), 寒冬(cold winter), 寒冷(cold), 旱灾(drought), 洪水(flood), 极冷(extremely cold), 季风(monsoon), 减排(emission reduction), 降水(precipitation), 降雪(snowfall), 降雨量(rainfall), 久旱(long drought), 飓风(hurricane), 空气污染(air pollution), 空气质量(air quality), 酷热(extremely hot), 酷暑(hot summer), 雷暴(thunderstorm), 雷击(lightning strike), 零排放(zero emission), 龙卷风(tornado), 隆冬(midwinter), 闷热(sultry), 暖冬(warm winter), 气候变化(climate change), 气候风险(climate risk), 气候灾难(climate catastrophe), 气温(air temperature), 气象(meteorological), 清洁发展(clean development), 取暖季(heating season), 全球

	变暖(global warming), 台风(typhoon), 碳捕捉(carbon capture), 碳储备(carbon stock), 碳排放(carbon emission), 碳强度(carbon intensity), 碳足迹(carbon footprint), 天气(weather), 温度异常(abnormal temperature), 温室气体(greenhouse gas), 雾霾(smog), 严冬(severe winter), 严寒(severe cold), 炎热(hot), 野火(wildfire), 厄尔尼诺(El Niño), 拉尼娜(La Nina), 气候(climate), 绿色转型(green transition), 转型升级(upgrade), 大气污染防治法(air pollution prevention and control law), 蓝天保卫战(blue sky defense), 柴油货车治理(diesel truck management), 长江保护(Yangtze River protection), 渤海综合治理(comprehensive treatment of Bohai Sea), 黑臭水体(black and smelly water), 绿色消费(green consumption), 绿色产品(green product), 新能源(new energy), 清洁生产(clean manufacturing), 绿色低碳(green and low carbon), 清洁能源(clean energy), 过剩产能(excess capacity), 低碳发展(low carbon development)
Uncertainty	不确定(uncertainty, uncertain), 不明确(unclear), 震荡(shock), 动荡(turmoil), 未明(unknown), 不明朗(not clear, unclear), 不清晰(not clear), 未清晰(not clear), 难以估计(hard to estimate), 无法预料(unpredictable), 无法预测(unpredictable), 无法预计(unpredictable), 无法估计(ineestimable), 不可预料(unexpected), 不可预测(not predictable), 不可预计(unpredictable), 波动(volatility), 不稳(unstable), 难以预计(unpredictable), 不可估计(ineestimable)
Policy	政策(policy), 制度(measures), 体制(system), 战略(strategy), 规章(regulations), 规例(regulation), 条例(regulations), 执政(govern), 政委(political commissar), 国务院(State Department), 人大(National People's Congress), 人民代表大会(People's Congress), 中央(central), 总书记(General Secretary), 国家领导人(national leader), 政治(politics), 政府(government, authority), 国家主席(president), 整改(rectification), 规管(regulation), 监管(supervision), 总理(prime minister), 改革(reform), 整治(regulation), 治理(governance), 统筹(overall planning), 协同(collaborate), 制定(formulate), 实施(implement), 相关部委(relevant ministries), 领导小组(leading group), 生态环境部(Ministry of Ecology and Environment), 文件(document), 方案(plan), 政策体系(policy system), 工作格局(work pattern), 协同治理(collaborative

governance), 政策法规 (policies and regulations), 试点示范 (pilot demonstration), 建立健全 (establish and improve), 产业结构调整 (industrial structure adjustment), 查处 (investigate), 依法依规 (according to laws and regulations), 专项检查 (special inspection), 清单管理 (list management), 动态监控 (dynamic monitoring), 通报批评 (bulletin of criticism), 监管体系 (regulatory system), 监督考核 (supervision and assessment), 评价指标 (evaluation indicators), 组织实施 (organization and implementation), 鼓励企业 (encourage business), 环境法规 (environmental regulations)

The classification of Chinese keywords about the China Climate Policy Uncertainty Index (CPU) is similar to Huang and Luk (2018), but the specific index method is different. Huang and Luk (2018) search the news that contains at least one keyword in these three groups. We made some improvements for a more accurate search. Specifically, we first download all articles that contain the vocabulary of uncertainty. In fact, many climate emergencies will lead to subsequent policy and administrative directives. Therefore, articles containing climate change-related words in their titles were further counted among the obtained articles. Then, many articles related to policies are not truly reflected in the news. For example, when the impact of a typhoon is relatively serious, the reports about the uncertain impact of the typhoon do not involve policy-related vocabulary, but the local government must issue relevant emergency measures to reduce the harm caused by such extreme events. Hence, finally, we conducted an investigation of the articles retrieved in the above process one by one and made a subjective judgment on whether there would be a response policy issued in the future for these articles, and many articles were deleted. Following Huang and Luk (2018), the sample starts in Jan 2000 and ends in Mar 2022.

Considering that the lag of policies and the number of relevant policies are relatively small, a monthly CPU will be adopted in the empirical analysis. The time trends for CPU are reported in Figure 1.

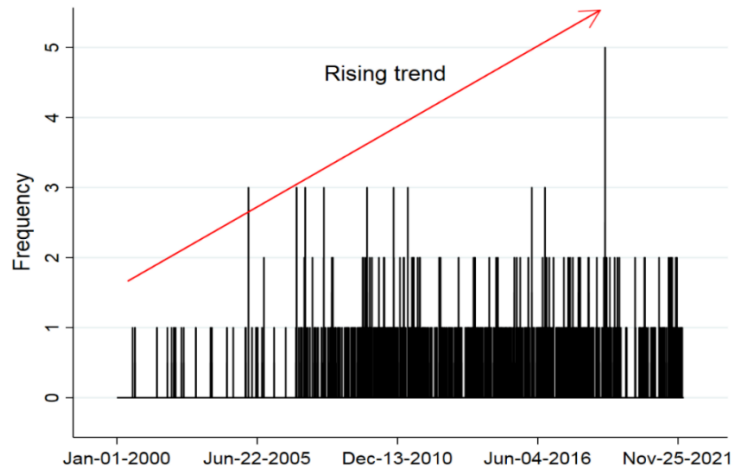


Figure 1 Time trend of China's daily CPU

As shown in Figure 1, the high level of the China's CPU is denser from 2005 to 2022, especially during the last decade, which indicates that the uncertainty of China's climate policy is increasing year by year, and climate change is attracting increasing attention from policy-makers. In this paper, considering the comparison with the condition of the US and the lag time of individual economic responses after policy releases is relatively long, we adopt the monthly China's CPU as the independent variable, which is consistent with the frequency of the variables in Huang and Luk (2018).

3.3 Measuring monthly volatility of Chinese and US stock index

This paper first calculated the log return (r) of the stock market index, and $r_t = \ln(P_t/P_{t-1})$, where P_t represents the closing price in period t . Then, the GARCH model is adopted to obtain the monthly volatility of the Chinese and US stock indices. Although ARCH could describe the volatility and obtain a very good effect, higher orders may be required in actual modeling. This paper adopts the GARCH (1,1) model, which is an important extension of the ARCH model. For a logarithmic series of returns r_t , we assume that its innovation $a_t = r_t - \mu_t = r_t - E(r_t|F_{t-1})$ and $\{a_t\}$ is GARCH(m,s) if $a_t = \sigma_t \varepsilon_t$ and $\sigma_t^2 = \alpha_0 + \sum_{i=1}^m \alpha_i a_{t-i}^2 + \sum_{j=1}^s \beta_j \sigma_{t-j}^2$, where $\{\varepsilon_t\}$ is independent identically distributed white noise sequences with zero mean unit variance. $\alpha_0 > 0$, $\alpha_i \geq 0$, $\beta_j \geq 0$, $0 < \sum_{i=1}^m \alpha_i + \sum_{j=1}^s \beta_j < 1$. The conditional variance can vary over time. The GARCH model has few grading methods and generally adopts the trial and error method.

Considering that GARCH (1,1) can meet the research needs in many cases, this paper adopts the GARCH (1,1) model to estimate the model volatility. The modeling process includes (1) testing the ARCH effect; (2) estimating the parametrics of the GARCH model and obtaining volatility; and (3) testing whether the model has extracted all the effective information, that is, whether the standardized residual conforms to the white noise sequence.

3.4 Measuring monthly correlation between Chinese and US stock index volatilities

To compare with linear correlation coefficient, rolling regression method with a 30-month window and Kendall's tau (τ) are adopted. τ is defined by

$$\tau = \Pr\{(X_1 - X_2)(Y_1 - Y_2) > 0\} - \Pr\{(X_1 - X_2)(Y_1 - Y_2) < 0\}$$

where $-1 \leq \tau \leq 1$. $\Pr\{(X_1 - X_2)(Y_1 - Y_2) > 0\}$ and $\Pr\{(X_1 - X_2)(Y_1 - Y_2) < 0\}$ represent the probabilities of concordance and discordance respectively. Similar to Pearson correlation coefficient, $\tau = -1$ and $\tau = 1$ represent perfect negative correlation and perfect positive correlation, respectively. When the copula function between X and Y is estimated, namely $C(u, v)$, Kendall's tau can be written as

$$\tau = 4 \int_0^1 \int_0^1 C(u, v) dC(u, v) - 1$$

where $u = F(X)$ and $v = F(Y)$.

3.5 Measuring upper tail dependence between Chinese and US stock volatilities

The impact of climate policy uncertainty on the upper tail dependences between Chinese and US stock market volatilities also attract interest. To be specific, tail correlation is when two financial markets both rise (upper tail correlation) or fall (lower tail correlation). This paper focus on the tail dependence of stock market volatilities instead of stock market price, and the main purpose is to analyze whether CPU could affect the correlation between China's and the US stock markets' volatilities under high level of volatility, namely upper tail dependence.

To depict this effect, copula is considered. The copula function can accurately construct the joint

distribution of multivariate random variables and then be used to calculate the tail correlation of multiple random variables. Specifically, the time series of upper tail dependence is calculated with the copula function and rolling window method. The width of the window is 30 and starts in Jun 2002 and ends in Mar 2022. Under each window, we use a bivariate copula to calculate the upper tail dependence. A bivariate copula $C(u, v)$ is a bivariate distribution function with standard uniform marginal distributions. $C(u, v) = P(U < u, V < v)$, where U and $V \sim U(0,1)$. In fact, function $C: [0,1]^2 \rightarrow [0,1]$. According to Sklar's Theorem, $H(x, y)$ is a bivariate distribution function with marginal distributions $F(x)$ and $G(y)$. There exists a copula $C: [0,1]^2 \rightarrow [0,1]$ such that $H(x, y) = C(F(x), G(y))$, and $(x, y) \in (-\infty, \infty)^2$. If $F(x)$ and $G(y)$ are continuous, C is unique. Considering that the distribution of these volatilities may not be a normal distribution, the most copula function should be selected first. Following Okhrin et al. (2021), we use goodness-of-fit tests to determine the copula, and after we find the copula, the upper tail dependence can be calculated. The upper tail dependence coefficient between X and Y is defined by

$$\lambda_U(X, Y) = P(U \geq u | V \geq v) = \lim_{u \rightarrow 0^+} \frac{\bar{C}(u, u)}{1 - u}$$

where $\bar{C}(1 - u, 1 - v) = 1 - u - v + C(u, v)$. If $\lambda_U(X, Y) \in (0, 1]$, X and Y have a upper tail dependence. If $\lambda_U(X, Y) = 0$, X and Y are independent in the upper tail.

3.6 The distribution lag nonlinear model (DLNM)

To quantitatively investigate the nonlinear and lag impacts of CPUs on stock market volatilities and their correlation, the DLNM method proposed by Gasparrini et al. (2010) is adopted. The DLNM method has been used widely in assessing the health effects of air pollution and temperature, and it also can be adopted to analyze the nonlinear and hysteretic relationships between variables. It can provide the different coefficients under different levels of CPU and different lag terms. According to Gasparrini et al. (2010), for example, a general model that is used to describe the nonlinear nexus between *CPU* and *Volatility* is as follows:

$$Volatility_t = \alpha + \sum_{j=1}^J s_j(CPU_{tj}; \beta_j) + \sum_{k=1}^K \gamma_k u_{tk} + \varepsilon_t \quad (3)$$

where u_{tk} represents control variables and has a linear impact on $Volatility_t$, and function s_j could represent the nonlinear impacts of *CPU* on *Volatility* when it is defined by a smooth

function, such as a spline function or polynomial function. The aim is to transform CPU_t into several time series with different coefficients that could depict its nonlinear impact. By using the smoothed functions, $s(CPU_t; \boldsymbol{\beta}) = \mathbf{z}_t^T \boldsymbol{\beta}$, the transformation can be shown in Eq. 1:

$$\begin{matrix} \mathbf{1} \\ CPU_1 \\ \vdots \\ CPU_t \\ \vdots \\ CPU_n \\ \mathbf{CPU} \end{matrix} \Rightarrow S \Rightarrow \begin{matrix} J = \mathbf{V}_x \\ \begin{bmatrix} Z_{11} & \cdots & Z_{1j} & \cdots & Z_{1J} \\ X_{t1} & \cdots & Z_{tj} & \cdots & Z_{tJ} \\ \vdots & & \vdots & & \vdots \\ X_{n1} & \cdots & Z_{nj} & \cdots & Z_{nJ} \end{bmatrix} \\ \mathbf{Z} \end{matrix} \quad (4)$$

where $\mathbf{z}_t$ is the t th row of the $n \times v_x$ basis matrix $\mathbf{Z}$. Then, the lag effect also can be introduced into the model. It is assumed that $Volatility_t$ could be affected by CPU_{t-l} , and l is the lag. The $n \times (L + 1)$ matrix $\mathbf{Q}$ can be obtained, in which:

$$\mathbf{q}_t = [CPU_t, \dots, CPU_{t-l}, \dots, CPU_{t-L}]^T \quad (5)$$

where L is the maximum lag and therefore $\mathbf{q}_1 \equiv \mathbf{CPU}$, which is the first column of $\mathbf{Q}$. Then, a new DLNM can be constructed and $s(CPU_t; \boldsymbol{\eta}) = \mathbf{q}_t^T \mathbf{C} \boldsymbol{\eta}$, where $\mathbf{C}$ denotes an $(L + 1) \times v_l$ matrix of basis variables obtained by adopting specific basis functions to the lag vector l . For example, if $\mathbf{C} \equiv \mathbf{1}$, it is a moving average model. $\boldsymbol{\eta}$ is the vector of parameters that should be estimated, and the real coefficients $\hat{\boldsymbol{\beta}} = \mathbf{C} \hat{\boldsymbol{\eta}}$.

4 Empirical analysis

4.1 Volatilities of China's and US stock markets

To obtain the monthly volatilities of the Chinese and US stock markets, the GARCH (1,1) model is employed. We first calculate the logarithmic returns of four indices (SSEC, NASDAQ, SCI, S&P500), and the results are provided in Figure 2.

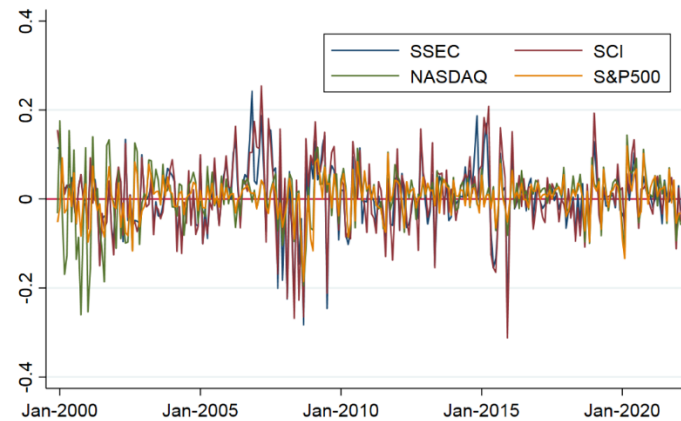


Figure 2 Time trend of log rate of returns of stock market index

As shown in Figure 2, the log rates of returns all fluctuate up and down near 0, showing a smooth character. Before adopting the GARCH model, the stationarity of the variables needs to be tested, and the lag order needs to be determined. The ADF test and PP test are used, and the null hypothesis is that the variable has a unit root. The results of unit root tests of monthly logarithmic returns are reported in Table 1.

Table 1 Unit root test of logarithmic returns

	SSEC	SCI	NASDAQ	S&P500
ADF test	-14.598***	-14.425***	-15.064***	-15.172***
	0.0000	0.0000	0.0000	0.0000
PP test	-14.762***	-14.526***	-15.036***	-15.176***
	0.0000	0.0000	0.0000	0.0000

As shown in Table 1, all p values are less than 0.01, indicating that all variables do not contain unit roots. Then, the autoregression models and Akaike information criterion (AIC) are adopted to select the order. The results are provided in *Supplementary Material: Table S1*. According to the results in *Supplementary Material: Table S1*, this paper confirms that both the SSEC and SCI have 4 lag orders and both the NASDAQ and S&P 500 have 1 lag order, and for the sake of the robustness of the conclusion, the lag orders for the NASDAQ and S&P 500 are determined to be 10. Then, the ordinary least squares (OLS) method is used, and the regression results are shown in *Supplementary Material: Table S2*. As shown in *Supplementary Material: Table S2*, the p values of the coefficients of the autoregression models are consistent with the AIC, and then the residuals are extracted and tested for the ARCH effect by using the LM test. The results are shown in *Supplementary Material:*

Table S3. From the results of LM tests, ARCH effects have been confirmed. Based on the above tests, this paper selected the mean regression models with 4 lag orders for the stock market index in China and with 1 lag order for the US. The results of GARCH (1,1) are reported in Table 2.

Table 2 Regression results of GARCH (1,1)

	SSEC	SCI	NASDAQ	S&P500
ARCH	0.1992*** 0.002	0.2119*** 0.019	0.2528*** 0.003	0.2378*** 0.000
GARCH	0.7495*** 0.000	0.7023*** 0.000	0.6893*** 0.000	0.7244*** 0.000
$\alpha + \beta$	0.9487	0.9142	0.9421	0.9622

Note: *, ** and *** indicates the coefficients are significant at 10%, 5% and 1% levels, respectively.

As shown in Table 2, the coefficients of residuals and conditional variances are significant at the 1% level, and their sums are less than 1, indicating that variances exist and are finite for the SSEC, SCI, NASDAQ and S&P 500. Then, the volatilities could be estimated and are reported in Figure 3.

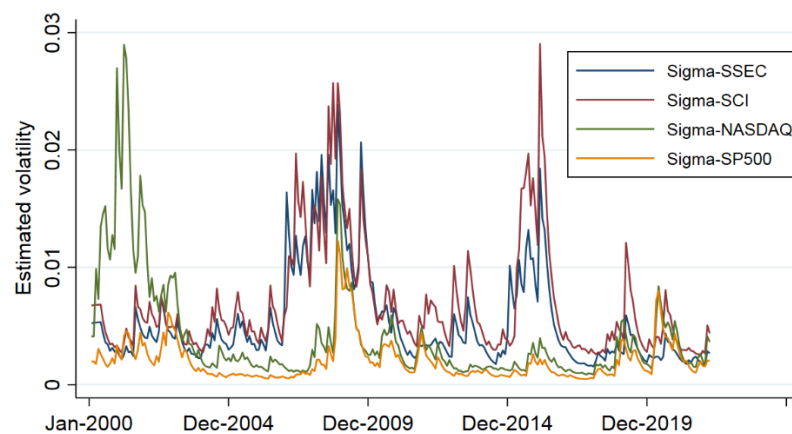


Figure 3 Estimated volatilities of different stock market index returns

As shown in Figure 3, stock market volatility was higher around 2001, 2008 and 2015. To ensure that the model has extracted useful information, the standardized residuals are examined by adopting the portmanteau test, and the results are reported in Table 3.

Table 3 Portmanteau test for white noise

	SSEC	SCI	NASDAQ	S&P500
(Q) statistic	36.9367	42.1690	33.4113	47.6932
P-values	0.6089	0.3773	0.7598	0.1884

According to Table 3, all p values are greater than 0.1, and the null hypothesis that the time series is white noise should be accepted. Therefore, it could be confirmed that all GARCH (1,1) models are reasonable.

4.2 Model construction and regression results

After calculating the CPUs for China and US and obtaining the volatilities, this paper aims to investigate the nonlinear and lag impacts of climate policy uncertainty on different stock market volatilities and their correlation for China and the US. The impacts on the tail dependence also are considered by using DLNM and copula function. We selected SSEC and NASDAQ to represent the stock market in China and the US, respectively, and the SCI and S&P 500 also are used in robustness tests for comparison. Specifically, after the unit root test, the empirical analysis can be divided into three parts: (1) GARCH models are adopted to obtain the volatility of the stock market in China and the US, and then the DLNM is used to discuss the nonlinear and lag impacts of the China's CPU and US CPU on stock market volatility. (2) The sliding time window method is used to calculate the time series of Pearson correlation coefficients between Chinese and US stock market volatility, and then the DLNM is adopted for empirical analysis. (3) The different goodness-of-fit (GoF) tests are employed to select the most common copula function to describe the dependency structure of stock market volatility and calculate their tail dependence with a sliding time window. The descriptive statistics of all variables and time trends are reported in Table 4 and Figure 4, respectively.

Table 4 Descriptive statistics

Variables	Obs	Mean	Std. Dev.	Min	Max
China's CPU	238	3.0966	2.8365	0	16
US CPU	238	111.6217	86.1953	4.43	629.02
σ of SSEC	238	0.0056	0.0044	0.0015	0.0239
σ of SCI	238	0.0071	0.0050	0.0024	0.0290
σ of NASDAQ	238	0.0029	0.0023	0.0009	0.0158

σ of S&P500	238	0.0020	0.0019	0.0005	0.0123
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Note: To be consistent with the correlation analysis, the samples reported in Table 4 were taken from June 2002 to March 2022 due to the use of the sliding window method, but in Figure 4, the time trends of CPU start since January 2000.

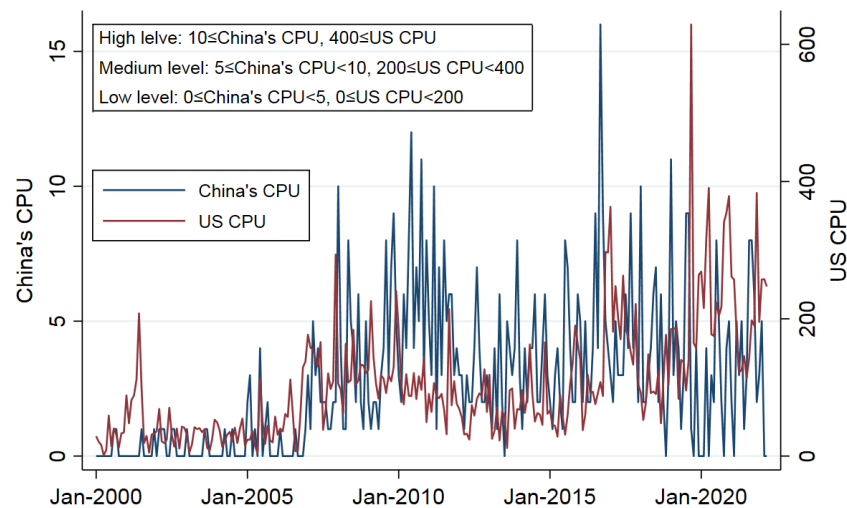


Figure 4 Time trends of China's CPU and US CPU

As shown in Table 4, the mean values and maximums of the China's CPU and US CPU have big differences, but the volatilities of the different stock indices are pretty close. Their time trends are provided in Figure 4 with different Y coordinate systems. During the period of 2000 to 2010, the climate policy uncertainties of China and the US were low, the overall level was rising, and volatility was more violent from 2010 to 2020. This indicates that climate change is receiving increasing policy attention from governments.

Table 5 Unit root test

	CPU_{China}	CPU_{US}	σ_{SSEC}	σ_{SCI}	σ_{NASDAQ}	$\sigma_{S\&P500}$
Dickey-Fuller (DF)	-10.299	-7.020	-3.828	-4.106	-3.975	-3.687
P-values	0.000	0.0000	0.0026	0.0009	0.0015	0.0043

From Table 5, according to the results of the unit root tests, the p values are less than 0.05, which rejects the null hypothesis of the unit root, and all variables are stationary time series.

4.2.1 The impacts of CPU on stock market returns

This paper first investigates the nonlinear and lag impacts of China's and US CPU on stock market returns from Jan 2000 to Mar 2022. These coefficients are reported in Figures 5 and 6.

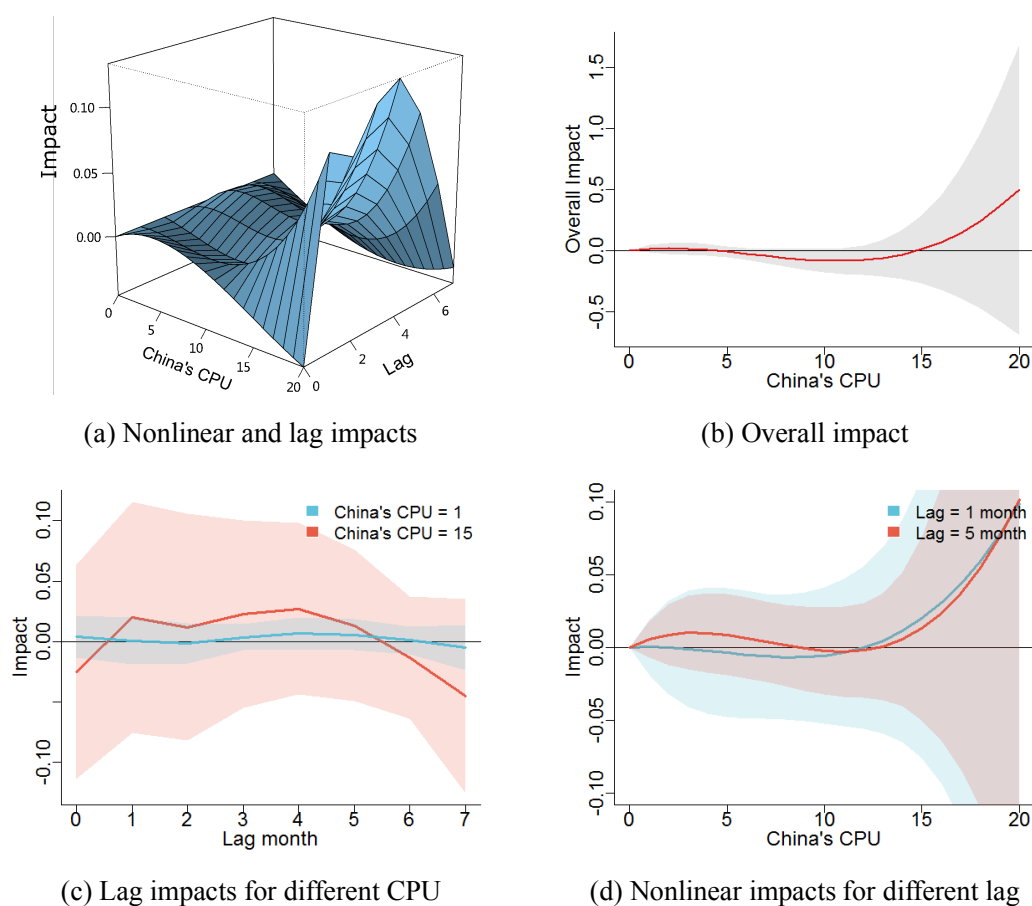


Figure 5 Impacts of China's CPU on SSEC return

As shown in Figure 5(a), China's CPU has significant nonlinear and lag impacts on SSEC return. The increase of current CPU has a negative influence on stock market return only under high CPU level, and this negative impact could be strengthened as the CPU continues to increase. When the CPU increases to about 20, the negative impact coefficient approaches -0.05. However, when the lag period is about 2 to 4 months, the increase of China's CPU has significant positive influence on stock market return also only under high CPU level. Then, when the lag exceeds 6 months, this positive effect on stock market return will turn negative again. From Figure 5(b), the cumulative effect of CPU on stock returns is significant and positive only at high CPU levels, which is due to

the superposition of influences from different periods. From Figure 5(c), different CPU has different lag impacts on stock market return. Only high level CPU could exert significant impact, and this effect will change from negative to positive and then to negative again with the increase of lag period. Figure 5(d) also indicates that only high level of CPU could have positive impact on stock market return.

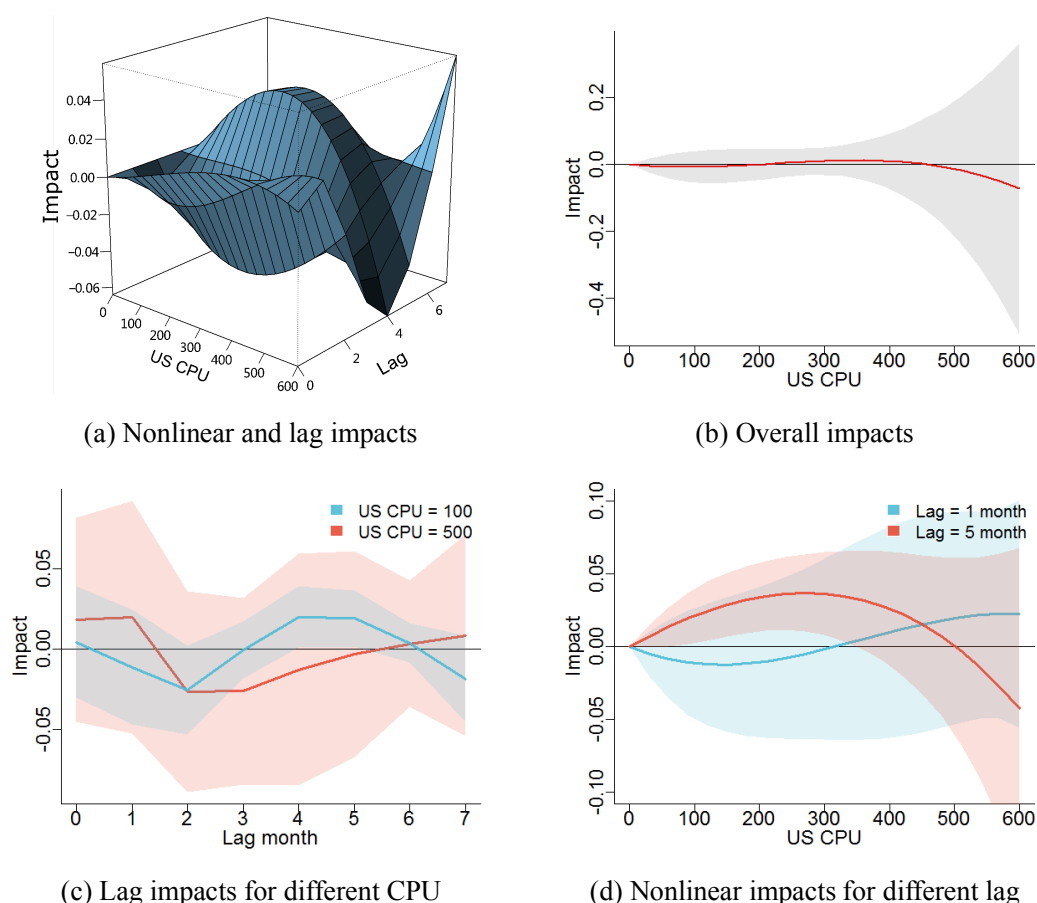


Figure 6 Impacts of US CPU on NASDAQ return

The impact of US CPU on stock market return is different from the impact of China's CPU. As shown in Figure 6(a), the increase of current US CPU does not have significant influence on stock market return under both low and high CPU level. From the perspective of lagged effects, the increase of current US CPU could have negative impact on the stock market return after one or two months. However, when the lag period is about 4 months, the increase of US CPU will have positive impact on the stock market return, and this impact could turn to be negative under high level of US CPU. Then, when the lag period reaches about 6 months, US CPU under high level

could have significant and positive impact on stock market return. From Figure 6(b), differing from China's CPU, US CPU under high level could have a negative overall impact on stock market return. Based on Figure 6(c), the impacts of US CPU under both low and high levels are similar, the impact is negative when the lag is about 2 months, but as the lag increases, the impact of US CPU on stock market return under level can be positive. Then it will turn to be negative after half a year. From Figure 6(d), the impacts of US CPU under different lag are different. Under higher lag period, US CPU is more likely to have a positive impact on stock market return, and this positive impact will turn to be negative under high level of CPU.

4.2.2 The impacts of CPU on stock market volatility

Except stock market return, the impacts of CPU on stock market volatility of China and the US are also investigated by DLMN method. The regression results are reported in Figures 7 and 8.

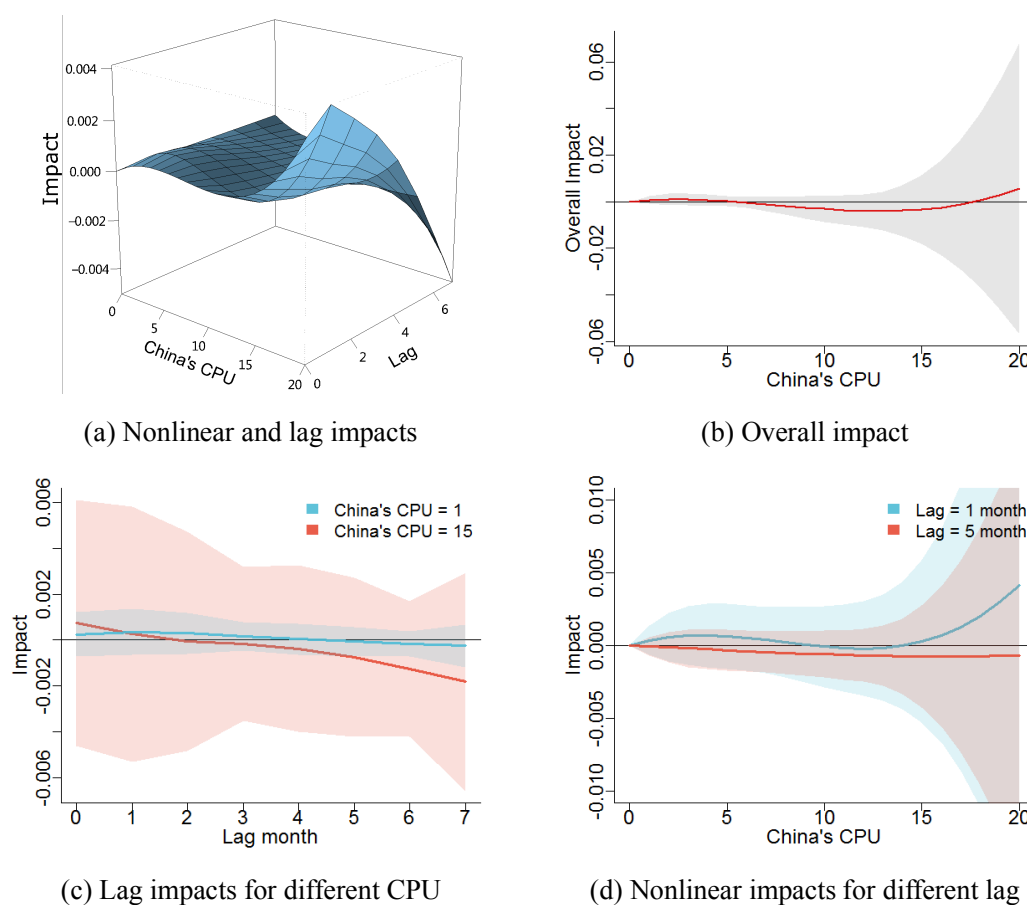
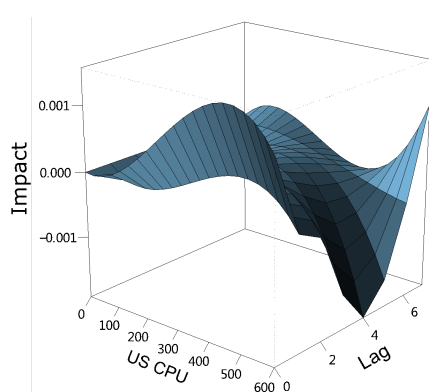
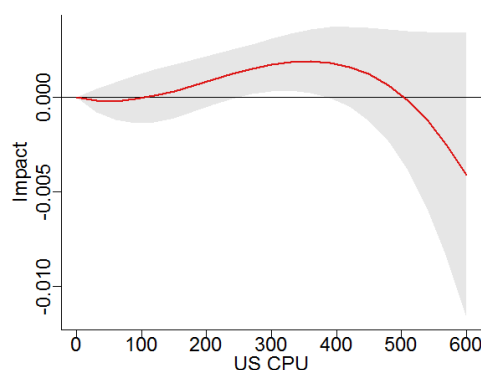


Figure 7 Impacts of China's CPU on SSEC volatility

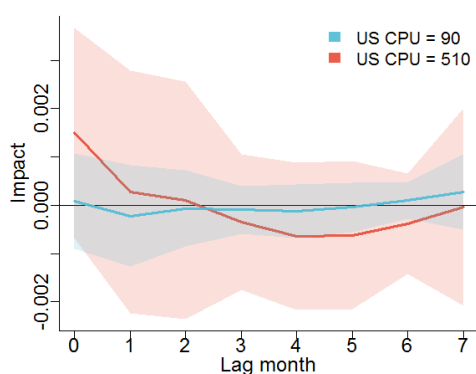
As shown in Figure 7(a), when the level of China's CPU is less than 15, the coefficients are close to 0, indicating that the impacts of the current period and lag periods are not obvious. When the China's CPU is greater than 15 and less than 20, the increase of the China's CPU from 15 to 20 will make the volatility of the SSEC decrease by 0.2%. The lag impacts also could increase with the change of time. It could be deduced that in China, when climate policy uncertainty is high, the negative impact on volatility will be strengthened and last for more than six months. The same condition also can be found in Figure 7(b), that the overall effect will turn to be positive when the China's CPU is greater than 15. Figure 7(c) provides the lag impacts of low and high China's CPUs for comparison. Only a high level of climate policy uncertainty could have a negative influence on SSEC volatility. Based on Figure 7(d), the nonlinear impacts for different lag are also different. China's CPU has a positive impact on stock market volatility only in the short term and under high level.



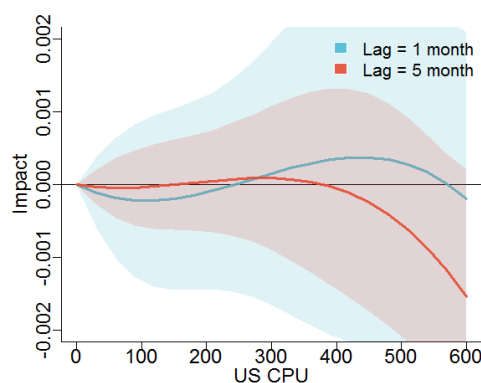
(a) Nonlinear and lag impacts



(b) Overall impact



(c) Lag impacts for different CPU



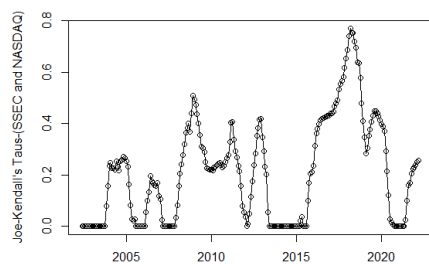
(d) Nonlinear impacts for different lag

Figure 8 Impacts of US CPU on NASDAQ volatility

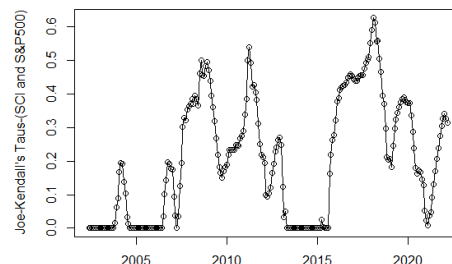
As shown in Figure 8, the nonlinear and lag impacts of the US CPU on the volatility of NASDAQ are different from those in China. From Figure 6(a), in the current period, the US CPU has a positive impact on the volatility of the NASDAQ. This positive impact could be strengthened with the increase of US CPU and reaches the maximum value when the US CPU equals about 400, then it will decrease to 0 as the US CPU increases to 600. Similar to China's CPU, US CPU under low level does not have significant impact on stock market volatility, but under high level of US CPU, the impact is different. When the lag period is about 4 months, US CPU under high level could have significant negative influence on stock market volatility, and when the lag exceeds about 6 months, this negative impact will turn to be positive. According to Figure 8(b), the overall effect of the US CPU is also different from that in China. With the increase of US CPU, the impact of US CPU on stock market volatility is significant and positive in the interval [200, 450] and turns to be negative when US CPU is under high level (US CPU is greater than 500). From Figure 8(c), as the lag increases, the impact of the US CPU under high level on stock market volatility is first positive and then turns to be negative. Based on Figure 8(d), the short-term impacts of US CPU are the opposite of the long-term impacts. Only when the lag is long the US CPU under high level could have a negative impact on stock market volatility. This indicates that it takes some time for US climate policy uncertainty to have a significant impact on stock market volatility, and when climate policy uncertainty is low, it has little impact on stock market volatility.

4.2.3 The impacts of CPU on Kendall's tau correlation between stock markets' volatilities

Kendall's tau is used to measure the correlation between stock markets' volatilities in China and the US, and this correlation coefficient is calculated by the rolling method. The rolling window is 30. The time trends of the SSEC and NASDAQ are reported in Figure 9, and the results of unit root tests are reported in Table 6.



(a) Correlation between SSEC and NASDAQ



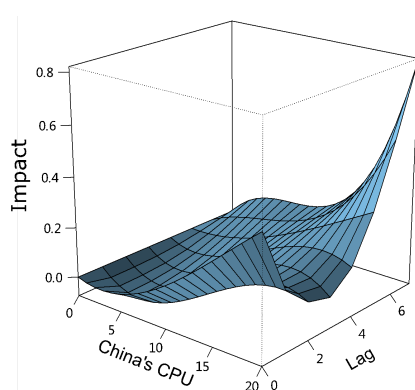
(a) Correlation between SCI and S&P500

Figure 9 Nonlinear correlation coefficients between volatility

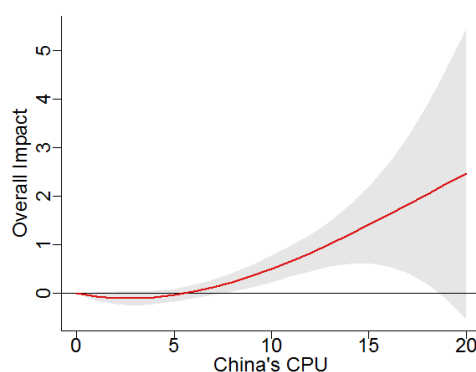
Table 6 Unit root tests of Kendall's tau (SSEC-NASDAQ and SCI-S&P500)

Unit root tests	Kendall's tau (SSEC-NASDAQ)	Kendall's tau (SCI-SP500)
Dickey-Fuller test	-1.602***	-1.661***
P-values	0.0553	0.0490

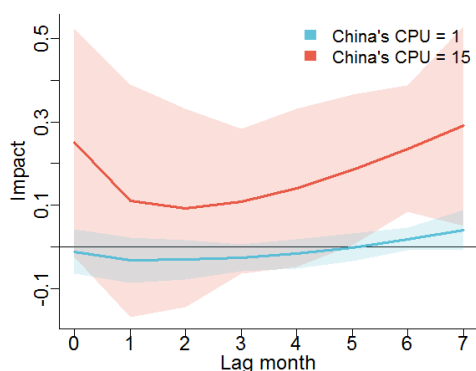
As shown in Figure 9, there is a significant positive nexus of the volatility between the SSEC and NASDAQ. During most times, the correlation is close to 0.4. According to Table 6, all p values are less than 0.01, indicating that the time series of correlations do not contain unit roots; therefore, the DLNM models could be used. The regression results are provided in Figures 10 and 11. Beyond that, the correlation between the SCI and S&P 500 will be used for the robustness test and the regression results are provided in *Supplementary material*.



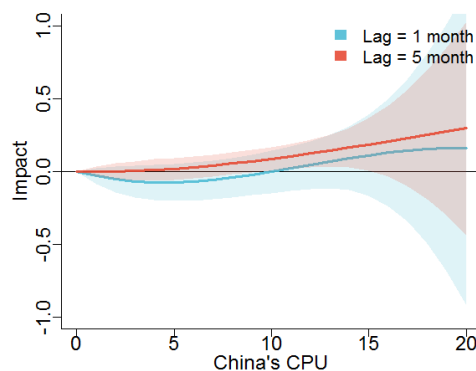
(a) Nonlinear and lag impacts



(b) Overall impact



(c) Lag impacts for different CPU



(d) Nonlinear impacts for different lag

Figure 10 Impacts of China's CPU on correlation (SSEC-NASDAQ)

As shown in Figure 10(a), in the current period, only when the China's CPU is under high level could the correlation between the SSEC and NASDAQ be positively affected. This positive impact can be strengthened with the increase of China's CPU. Beyond that, the positive impact of China's CPU under high level (China's CPU is greater than 15) could be weakened and then strengthened as the lag increase. From Figure 10(b), when China's CPU is greater than 6, the overall impact of China's CPU on correlation is positive and significant, and this positive impact will be strengthened with the increase of China's CPU. As shown in Figure 10(c), China's CPU under high level, such as China's CPU equals to 15, could have a positive influence on the correlation between stock market's volatilities, and this positive impact will be weakened and then strengthened as the lag increases. From Figure 10(d), the impact of China's CPU on correlation is not significant under low level, but could be positive under high level. On the whole, China's climate policy uncertainty has a positive impact on the correlation of stock market volatility only when it is high, and the impact will reach its maximum value half a year later.

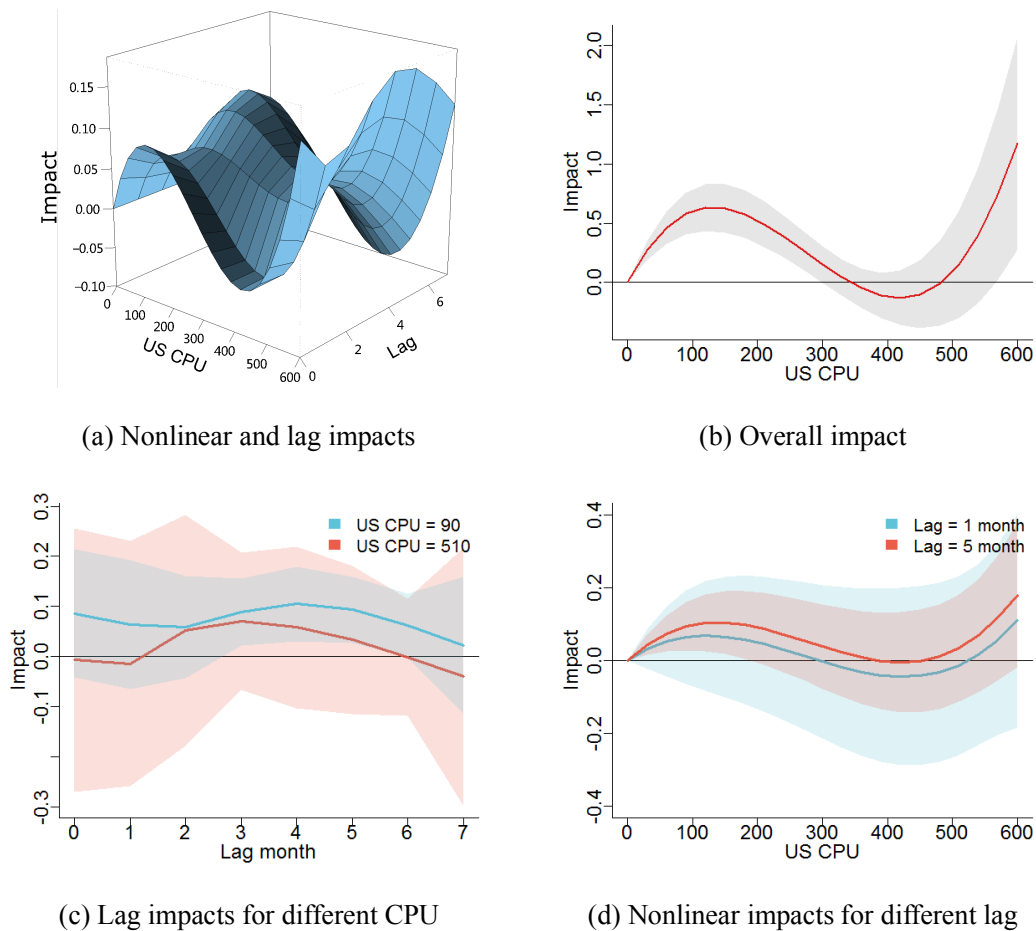


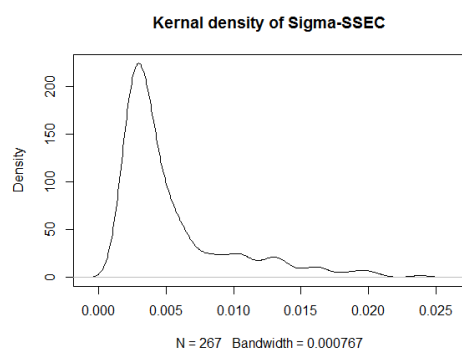
Figure 11 Impacts of US CPU on correlation (SSEC-NASDAQ)

As shown in Figure 11, the impacts of US CPU on correlation are more complex than that of China's CPU. From Figure 11(a), in current period, the impact of US CPU is nonlinear and N-shaped, when US CPU is in the interval $[0, 200]$, the impact is positive and reaches the maximum when US CPU equals to 100. This impact will turn to be negative when US CPU is in the interval $[200, 500]$, and it will reaches the minimum when US CPU equals to 400. As US CPU increase, US CPU under high level could have a positive impact on correlation again. From Figure 11(b), the overall impact of US CPU is significant, nonlinear and N-shaped. When US CPU is less than 300, it can have a positive cumulative impact on correlation, and this impact will reach maximum when US CPU equals to 150. However, with the increase of US CPU, this impact could turn to be negative when US CPU is greater than 350 and turn into positive again if US CPU is greater than 500. It could be further strengthened under high level of US CPU. From Figure 11(c), the impacts of US CPU under low level and high level are similar, which are positive when the lag period is in the interval $[2, 6]$, Figure 11(d) also confirms the similarity between the impacts of US CPU under low level and high

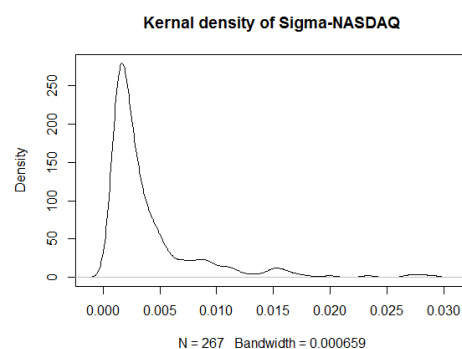
level. The short-term impact of US CPU which is in the interval $[350,500]$ on correlation could be negative. In summary, when the uncertainty of US climate policy is low, the correlation between Chinese and US stock market volatility can be improved. With the increase of climate policy uncertainty, this positive effect will become negative. However, the effect turns positive again when climate policy uncertainty is high. The same situation exists in different lag periods.

4.2.4 The impacts of CPU on upper tail dependence between stock markets' volatilities

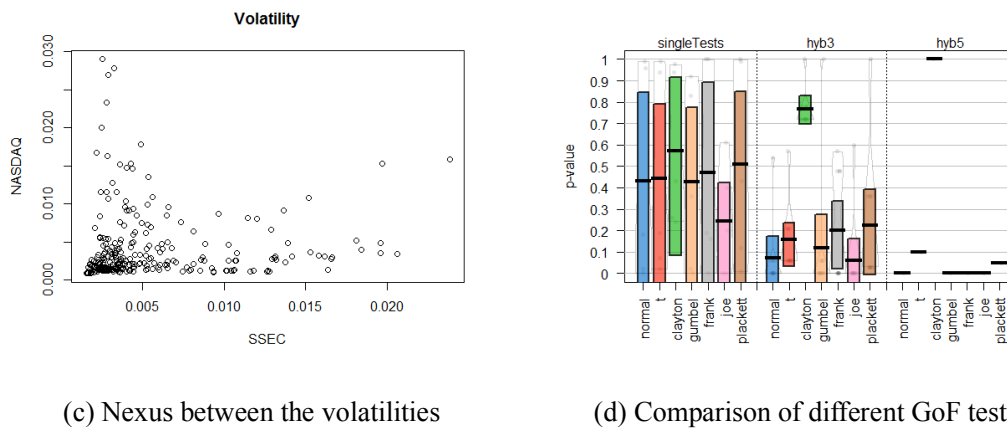
In addition to the discussion about the complex impacts of CPU on the correlation between the stock markets of China and the US, the impact on their tail dependence also is considered, especially the upper tail dependence, which has an important impact on the stability of financial markets and the normal operation of financial institutions, and investigating this point could help investors avoid financial investment risks from the perspective of climate policy and economic transformation. We first report the distribution of each volatility and its nexus. Then, different goodness-of-fit (GoF) tests are used to select the most suitable copula to calculate both lower and upper tail dependence. The nexus between the SCI and S&P 500 is also adopted as a robustness test. Then, we conduct unit root tests and use the DLNM for empirical analysis. The kernel densities of the SSEC and NASDAQ and the selection of the copula between these two indices are reported in Figure 12.



(a) Distribution of volatility of SSEC



(b) Distribution of volatility of NASDAQ



(c) Nexus between the volatilities (d) Comparison of different GoF tests

Figure 12 Distribution and GoF test for the volatilities (SSEC and NASDAQ)

As shown in Figures 12(a) and 12(b), both distributions of the volatilities of the SSEC and NASDAQ are not Gaussian distributions; therefore, the bivariate normal distribution function is inappropriate. From Figure 12(d), based on the results of different GoF tests, the most suitable copula is clayton for the upper tail dependence between the volatilities of the SSEC and NASDAQ. Then, we estimate the parametric of these copula and calculate the upper dependence. The rolling window method is adopted with a width equal to 30. Therefore, this time series starts in Jun 2002 and ends in Mar 2022.

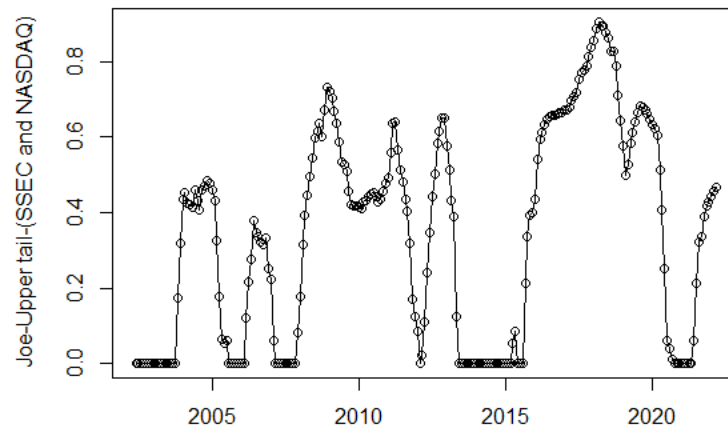


Figure 13 Time trend of upper tail dependence between SSEC and NASDAQ

Table 7 Unit root tests for tail dependence of SSEC-NASDAQ

Unit root tests	Upper tail Dependence
Dickey-Fuller test	-1.761**
P-value	0.0398

As shown in Figure 13 and Table 7, all p values are less than 0.05, indicating that the time series of tail dependence does not contain a unit root. Therefore, the DLNM method could be used and all results are provided in Figures 14 and 15.

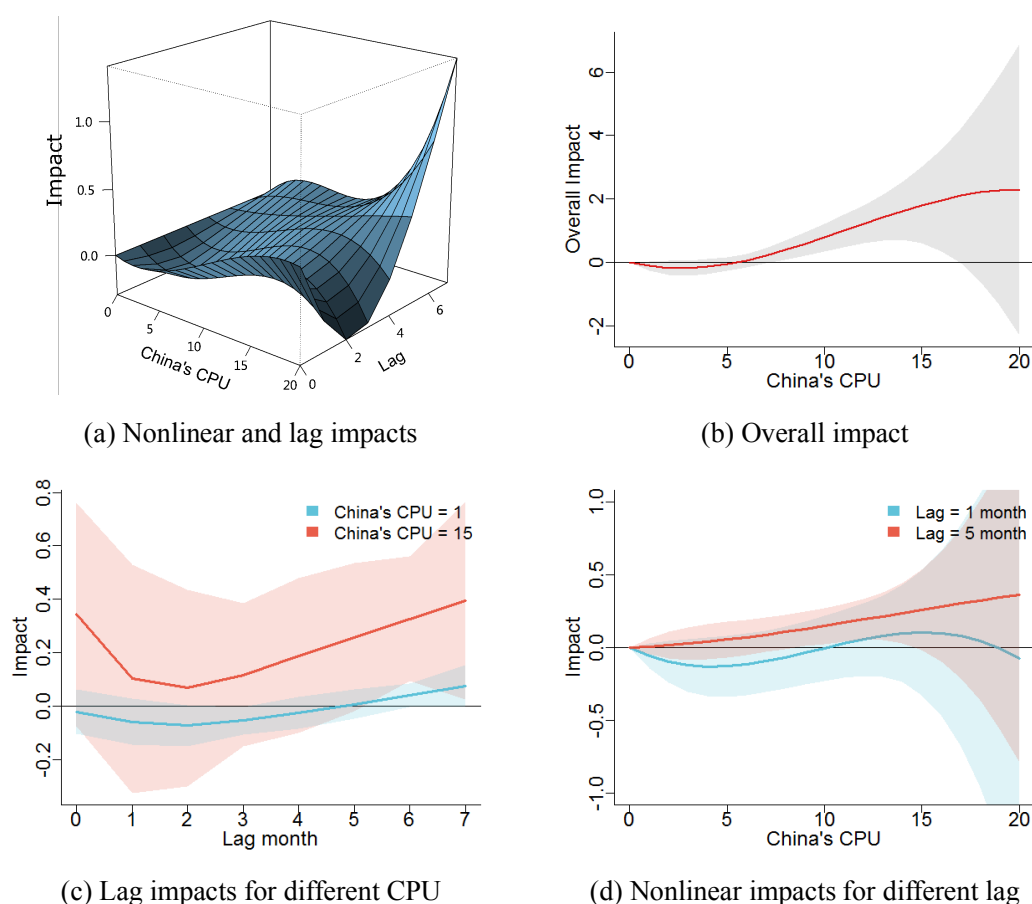


Figure 14 Impacts of China's CPU on upper tail dependence (SSEC-NASDAQ)

Upper tail dependence represents the correlation between two stock markets when volatility is very high. As shown in Figure 14(a), the current China's CPU under a high level could have a positive impact on upper tail dependence. This positive impact could turn into negative in two or three months. When the lag exceeds 4 months, the negative effect will become positive and be enhanced with the increase of lag. However, the impact of China's CPU under a low level on upper tail dependence is not significant, and even if the lag is increasing, there is no significant effect. From Figure 14(b), when China's CPU is greater than 6, it will have a significant positive overall impact on upper tail dependence, and this cumulative effect will be strengthened with the increase of

China's CPU. From Figure 14(c), only when China's CPU is under a high level could it have a positive impact on upper tail dependence, and this positive impact may be strengthened with the increase of lag month. Based on Figure 14(d), long-term China's CPU is more likely to have a positive effect on the upper tail dependence. In summary, China's climate policy uncertainty has an impact on upper tail dependence only at a high level, and this impact will change with the increase of lag time. The positive effect will be enhanced after more than 4 months of lag, that is, the upper tail dependence will be significantly increased.

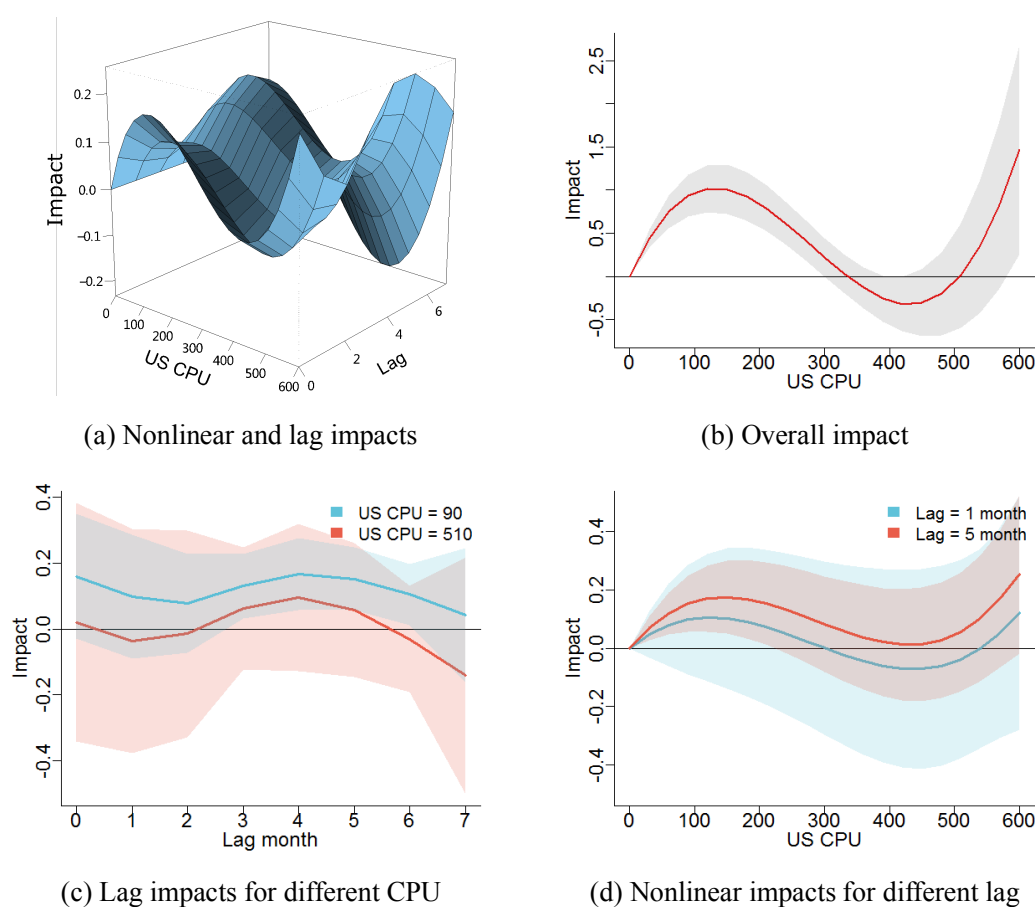


Figure 15 Impacts of US CPU on upper tail dependence (SSEC-NASDAQ)

As shown in Figure 15, the impacts of US CPU on upper tail dependence are more complex than that of China's CPU. From Figure 15(a), US CPU under low level has a positive impact on upper tail dependence, and this positive impact could reach the maximum when US CPU approaches 150. Then, it will be weakened with the increase of US CPU. However, US CPU under a high level will have a significant positive impact again. With the increase of the lag period, the positive impact of

US CPU at low level on the stock market will be weakened first, and then strengthened under about 4 lag months. Beyond that, the impact of US CPU under high level on upper tail dependence could be further strengthened as time goes on, and it will reach the maximum after six months. From Figure 15(b), the overall impact of US CPU is significant, nonlinear and N-shaped. When US CPU is less than 300, it can have a positive cumulative impact on upper tail dependence, and this impact will reach maximum when US CPU equals to 150. However, with the increase of US CPU, this impact could turn to be negative when US CPU is greater than 350 and turn into positive again if US CPU is greater than about 520, and it could be further strengthened under high level of US CPU. This is similar to the impact of US CPU on the correlation between stock markets' volatilities. From Figure 15(c), the lag impact of US CPU under a low level on upper tail dependence is positive, and from Figure 15(d), with the increase of US CPU, the short impact of US CPU could turn from positive into negative and then turn into positive again. In summary, when the uncertainty of US climate policy is low, it has a positive influence on the tail dependence of the current stock market, and this positive influence will be enhanced with the increase of lag period. When the uncertainty of US climate policy is between 400 and 500, this positive effect may be reduced to 0, and is not significant at different lag times. When climate policy uncertainty is very high, this positive effect is reinforced again and reaches its maximum at a lag of about 4 to 5 months.

The impacts of CPU on SCI, S&P500 and their correlation are provided in *Supplementary Material* as robustness test. Compared with these results, the impact are similar and the robustness of above conclusions are confirmed.

5 Conclusions

This paper calculates the daily and monthly indices of climate policy uncertainty in China for the first time from January 2000 to March 2022. The calculation process is based on the textual analysis method and refers to the methods of the climate policy uncertainty index and economic policy uncertainty index of the US. Then, according to this index and the US CPU, we use the DLNM model to compare and analyze the nonlinear and lagged effects of the CPU on stock market volatility, volatility correlation and upper tail dependence of volatilities in China and the US. We not only find

that the time trends of CPU indices in China and the US from 2000 to 2022 are very similar but also confirm the significant differences between the nonlinear and lag impacts of these two CPU indices on stock markets.

First, the magnitude of climate policy uncertainty in China and the US is different due to the different news media and text criteria selected in the statistical process, but they show a similar upward trend from January 2000 to March 2022. According to our assessment, China's climate policy uncertainty was low from 2000 to 2006, a period when climate change and its impacts were not widely considered by the government or residents. In 2009, the 15th Conference of the Parties (COP 15) to the United Nations Framework Convention on Climate Change (UNFCCC) was held in Copenhagen, Denmark, letting climate change and low-carbon development once again become a priority for all countries. China has actively formulated and implemented a series of climate change strategies, regulations and policies. For example, the Chinese government has formulated and released implementation plans for peaking carbon emissions in energy, industry, urban and rural development, transportation, agriculture, rural areas and other sectors and actively planned measures for carbon sequestration, energy transformation, pollution reduction and carbon reduction. Therefore, the uncertainty of China's climate policy began to rise in approximately 2009, and the high level of climate policy uncertainty has continued until now.

Second, for China: (1) Low climate policy uncertainty does not affect stock market return, but high climate policy uncertainty could decrease current stock market return, and this negative impact will be strengthened as China's CPU increase. Short-term impact of climate policy uncertainty is positive, but in the long term climate policy uncertainty could decrease stock market return. (2) High climate policy uncertainty increase current stock market volatility but decrease it in the future. (3) High climate policy uncertainty could increase the correlation between China's and the US stock markets' volatilities, and this positive impact will be strengthened in the long term. (4) High climate policy uncertainty could increase the upper tail dependence between China's and the US stock markets' volatilities in current period, decrease it after 2 months and increase it again after 5 months.

Third, for the US: (1) Climate policy uncertainty could decrease stock market return in the short

term but increase it in the long term. (2) High climate policy uncertainty could increase stock market volatility in short term, decrease volatility in 5 months and increase it again after 6 months. (3) Both low and high climate policy uncertainty could increase the correlation between China's and US stock markets' volatilities, and these positive could last for up to six months. (4) Low and high climate policy uncertainty could increase the upper tail dependence between China's and US stock markets' volatilities, but climate policy uncertainty under medium level have a negative impact on it.

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